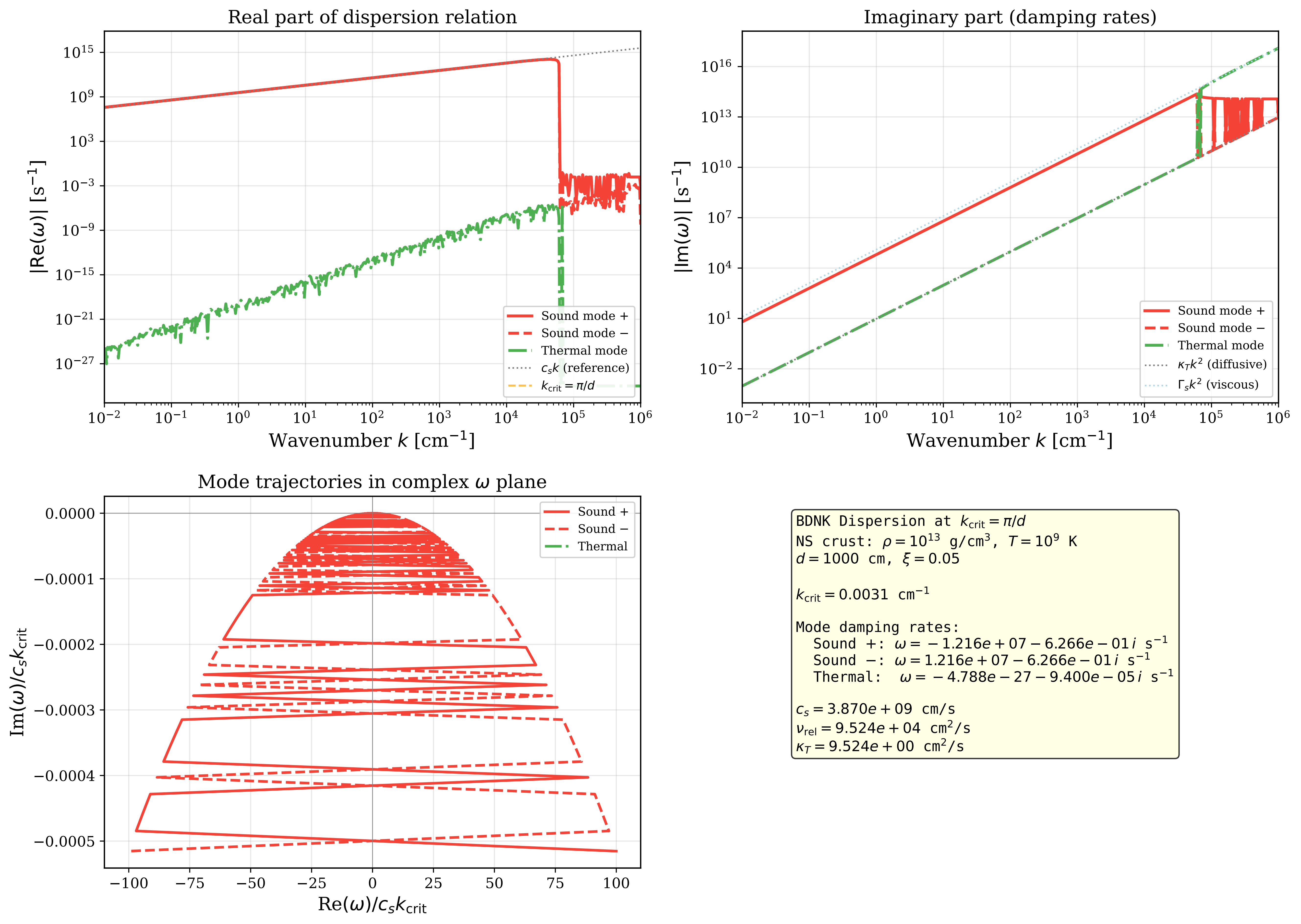

Real part of dispersion relation
|Re(ω)| [s^{-1}]
Wavenumber k [cm^{-1}]
Sound mode +
Sound mode −
Thermal mode
$c_s k$ (reference)
$k_{\rm crit} = \pi/d$
Imaginary part (damping rates)
|Im(ω)| [s^{-1}]
Wavenumber k [cm^{-1}]
Sound mode +
Sound mode −
Thermal mode
$\kappa_T k^2$ (diffusive)
$\Gamma_s k^2$ (viscous)
Mode trajectories in complex ω plane
Im(ω)/$c_s k_{\rm crit}$
Re(ω)/$c_s k_{\rm crit}$
Sound +
Sound −
Thermal
BDNK Dispersion at $k_{\rm crit} = \pi/d$
NS crust: $\rho = 10^{13}$ g/cm^3, $T = 10^9$ K
$d = 1000$ cm, $\xi = 0.05$
$k_{\rm crit} = 0.0031$ cm^{-1}
Mode damping rates:
Sound +: $\omega = -1.216e+07 - 6.266e-01\,i$ s^{-1}
Sound −: $\omega = 1.216e+07 - 6.266e-01\,i$ s^{-1}
Thermal: $\omega = -4.788e-27 - 9.400e-05\,i$ s^{-1}
$c_s = 3.870e+09$ cm/s
$\nu_{\rm rel} = 9.524e+04$ cm^2/s
$\kappa_T = 9.524e+00$ cm^2/s